

It is RECOMMENDED that a digest algorithm be chosen that has a minimum digest length of 256 bits, such as **sha-256** (ID 1 in the registry).

ImageDigest field

The ImageDigestField SHALL contain the digest of the entire payload of length **PayloadSize** that follows the header. The digest SHALL be computed using the algorithm indicated in the **ImageDigest-Type** field. This digest SHOULD be used by OTA Providers to ensure they have obtained the entire image expected, and that the contents matches the expectations.

11.21.3. Security considerations

The OTA Software Image file format does not specify the mechanisms that an OTA Requestor should use to validate that a software image is valid for itself. Considerations relative to OTA Software Image Signing are presented in [Section 11.20.4.2, “Image Verification”](#).

11.22. Bulk Data Exchange Protocol (BDX)

11.22.1. Overview

Nodes need the ability to transfer "files" between nodes. For example, uploading sensor data or diagnostics data to another node or downloading software update images from a software update server both require such a protocol.

This document defines a Bulk Data Exchange (BDX) protocol, where files are modeled as collections of bytes with some attached metadata. For the purposes of this protocol, files are opaque, and no attempt is made to specify their format. However, the protocol allows for extensible metadata so that higher-level applications can participate in the decision whether to proceed with a requested file transfer.

The Bulk Data Exchange (BDX) protocol has some semantic elements influenced by the Trivial File Transfer Protocol (TFTP) [RFC 1350](#).

One major difference is that TFTP is defined to run over UDP only, while Bulk Data Transfers can proceed over various reliable transports including both TCP and UDP, using the Message Reliability Protocol (see [Section 4.12, “Message Reliability Protocol \(MRP\)”](#)). Therefore, **BlockAck** and **BlockQuery** fulfill the role of application-layer control flow and acknowledgement rather than providing a reliability and retransmission mechanism. The availability of application-level flow control enables highly power-constrained nodes to pace transfers in a way that respects their power limitations.

11.22.2. Terminology

11.22.2.1. BDX Sender

The node that has bulk data to send to another node.

11.22.2.2. BDX Receiver

The node that receives bulk data from a Sender.